

AI DELIVERY & VALUE REALIZATION

A Predictive Supply Chain & Sales Forecasting Framework

From data pipeline to measurable business value — a reusable, end-to-end approach to AI-driven delivery risk, ETA, and demand forecasting

One reusable framework, four connected disciplines

We built and proved out an end-to-end methodology that turns raw logistics and sales data into predictive models — and turns those models into tracked business value.

1

What we built

An end-to-end ML pipeline: ingestion, feature engineering, model training, and evaluation for three predictive use cases.

2

What we proved

The full framework — technical, value-tracking, governance, and cloud scaling — works together as a repeatable delivery pattern.

3

Why it matters

A structured path from pilot to production: connect real data, and the same framework scales across regions and product lines.

Supply chains generate data — most of it isn't used predictively

THE CHALLENGE

- Late deliveries are discovered after they happen, not before
- Inventory and production decisions rely on reactive, backward-looking reports
- Sales forecasts are inconsistent across products and regions
- No shared way to translate operational data into quantified business value

THE OPPORTUNITY

- Predict delivery risk and ETA before disruption reaches the customer
- Move inventory and production planning from reactive to proactive
- Standardize forecasting across products/regions on one framework
- Track ROI, efficiency, and risk mitigation on a single value dashboard

Four connected pillars, one delivery model

1

Technical Core

Predictive ML models for Sales Forecasting, Late Delivery Risk, and Estimated Time of Delivery, built on a shared feature pipeline.

2

Value Realization Dashboard

KPIs that translate model output into business terms: Simulated ROI, Fulfillment Efficiency, and Risk Mitigation.

3

Delivery & Governance Artifacts

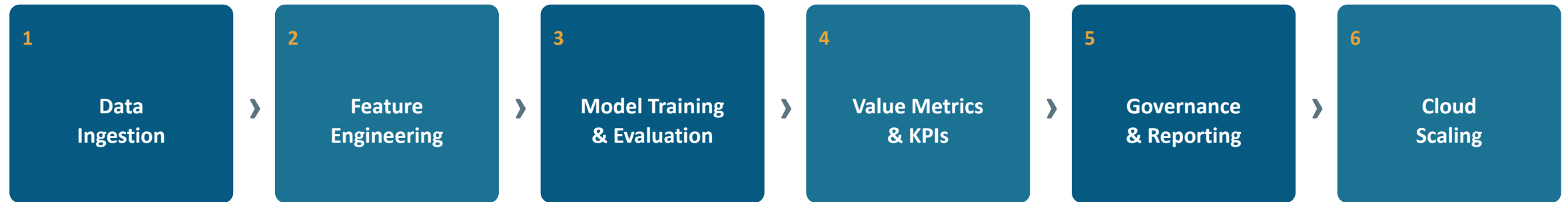
Kickoff documentation, an 11-week workflow timeline, and an executive reporting cadence for cross-functional alignment.

4

Cloud & Architecture Alignment

A blueprint for scaling ingestion, model execution, and automated reporting across a global logistics network.

A single pipeline feeds every model and every metric



Python + T-SQL data pipeline • Random Forest models (classification & regression) • Coca-Cola-style supply chain dataset used as proof-of-concept

Why this matters: Because every model shares the same ingestion and feature layer, adding a new use case — or a new region — means extending the pipeline once, not rebuilding it for each model.

Three predictive models, one shared pipeline

Late Delivery Risk

Classification (Random Forest)

98%

Recall on late shipments

78% accuracy • 79% precision • 0.88 F1

Sales Forecasting

Regression (Random Forest)

POC

Proof-of-concept only

MAPE 129% on synthetic data — signal-limited by design

Estimated Time of Delivery

Regression (Random Forest)

POC

Proof-of-concept only

MSE 1.52 days — needs real operational features

Results shown are from a proof-of-concept run on synthetic supply chain data. The Late Delivery Risk model demonstrates the framework's ability to flag risk early; forecasting accuracy is expected to improve substantially once trained on real operational data.

Turning model output into a shared language for value

\$67,300

Simulated ROI

Net avoided cost from proactively flagged late deliveries

98.35%

Risk Mitigation (Recall)

Share of true late deliveries caught before they happen

2.37%

Fulfillment Efficiency

On-time precision — the metric to improve first with real data

These figures are computed on synthetic proof-of-concept data to validate the metric definitions and dashboard logic — not to represent production-scale returns. The same formulas apply directly once live data is connected.

An 11-week path from kickoff to handover



Governance cadence: Kickoff document, workflow timeline, and executive presentation are maintained as living artifacts through weekly syncs and phase-end milestone reviews.

Built to scale into a global logistics network

Data Ingestion

Streaming + batch from ERP, TMS, WMS, CRM and market feeds

Processing & Storage

Data lake, warehouse, ETL/ELT orchestration, feature store

Model Execution

Managed training, endpoints for real-time & batch scoring, drift monitoring

Reporting & Governance

BI dashboards, automated alerts, access control & audit logging

Cloud-agnostic blueprint (GCP / AWS / Azure equivalents) — supports continuous ingestion and automated reporting without manual intervention.

From proof-of-concept to production pilot

1

Connect production data

Replace synthetic data with live ERP/TMS/CRM feeds to validate real-world model accuracy.

2

Pilot deployment

Run the Late Delivery Risk model in one region as a live pilot with operational teams.

3

Expand coverage

Extend the same pipeline to additional regions and product lines once the pilot validates value.

4

Invest in forecasting depth

Add advanced time-series methods to lift Sales Forecasting and ETA accuracy beyond proof-of-concept.

Thank you

Discussion & Q&A